

CSS-Native Layout Manager Current Status Report

Report Date: 2026-05-21 | **Scope:** NEW0 through NEW3 Batch 5 | **Evidence Base:** Tracked code, canonical diagnostics, runtime proof, stress corpus v1

Evidence Currency Legend

CURRENT

Reflects state at 2026-05-21 with verified canonical measurement.

HISTORICAL

Reflects state at a prior batch; preserved for trail; not current.

STRESS-ONLY

Measured against synthetic stress corpus; does not predict production behavior directly.

PROPOSAL

Awaiting user decision; not implemented.

SUPERSEDED

Replaced by a later finding; documented for completeness.

One-Paragraph Executive Summary

Runtime integration is promising; static visual quality is improving; production promotion remains blocked by contract and visual-polish decisions. The runtime dispatch path is proven stable (Lane B: 7/7 PASS on interaction stress, no DOM leak, correct state propagation) **CURRENT** . The visual diagnostics gate hard-fail free (0 clipped artwork, 0 off-page, 0 SVG overlap) across production templates **CURRENT** . Stress testing on 100 generated scenes shows the layout engine's CSS dispatcher executes reliably but synthetic scenes reveal much higher failure rates than production **STRESS-ONLY** . A hard-fail rule against cropped scientific assets was successfully operationalized (Workstream N: 3/3 recovery, no regression) **CURRENT** . However, PRIMARY_CONTRACT.md item 3 (layout engine ownership of object positioning) remains unresolved by design — NEW0 bypasses the layout engine entirely. Visual quality across scenes is 0 PASS / 4 PASS_TEMPLATE / 6 WARN on canonical precheck, with the six composition scenes warning on label readability and supporting-object proximity rather than primary-object visibility **CURRENT** . The scorecard gap for hard-fail detection remains the highest near-term risk. Recommendation: Accept NEW0 as a stable layer-1

runtime foundation, continue NEW2 work (no-crop polish), but defer production migration to pending resolution of contract and visual-polish decisions.

1. Executive Summary

The CSS-native layout system has reached a stable state across three distinct evidence layers. The runtime dispatch path is proven: a real click on a scene object flows through the validator, ObjectStateChange handler, and CSS-native adapter re-render cycle with zero DOM leaks or state corruption. The static-template visual gate is clean: zero hard-fail diagnostics (clipped artwork, off-page, SVG overlap, region overflow) across the 10 production templates. The diagnostic tools themselves (precheck, scorecard, artwork-integrity) are well-instrumented and detect failure modes accurately.

However, two significant gaps remain:

- **Visual quality is uneven.** Composition scenes warn on label readability and supporting-object proximity (6 of 10 scenes WARN on secondary factors, not primary-object visibility). The layout is compositionally functional but not pedagogically polished.
- **The design contract is unresolved.** PRIMARY_CONTRACT.md item 3 states that scene object layout must be handled by the layout engine. NEW0 bypasses the engine entirely; objects are positioned by CSS Grid and Flexbox rules, not by YAML-declared primitives. Production promotion would require either amending the contract or integrating NEW0 as the layout engine's output backend.

Recommendation: Move forward with NEW2 visual polish and NEW3 stress-corpus work (no-crop rule enforcement, scorecard alignment). Plan a user decision on PRIMARY_CONTRACT.md amendment in parallel. Production promotion remains blocked pending contract resolution and visual-polish decisions.

2. Layer Model

Evidence in this report belongs to one of three independent, stackable layers. Each layer carries its own verification step, supports its own next-step workstream, and does not imply success in the other layers.

Layer 1: Runtime Dispatch (TypeScript, production bundle)

A user interaction (click on a scene object) flows through:

```
click-handler validator preset ObjectStateChange renderScene re-invoke CSS-native adapter
```

Verification: Lane B interaction stress test on the built bundle. Result: 7 of 7 PASS on click target binding, state change application, and re-render proof (invocation count delta +1, DOM children unchanged).

Why separate: Runtime success does not guarantee visual quality or hard-fail cleanliness. It only proves the dispatch path executes without error.

Layer 2: Static Visual Quality (CSS + HTML templates, precheck diagnostics)

The rendered scene meets spatial, proportional, and aesthetic goals: primary objects visibly dominant, supporting objects nearby, labels readable, no artwork clipping.

Verification: Lane A precheck contact sheets and scorecard. Result: 0 hard fails (PASS gate), 6 composition scenes WARN on secondary factors (label overlap, supporting-object distance), 4 template scenes PASS_TEMPLATE.

Why separate: Visual quality is independent of runtime. A scene can render correctly and still have cropped artwork or illegible labels.

Layer 3: Diagnostic Integrity (Measurement tools, guards against metric gaming)

The precheck, scorecard, and artwork-integrity tools must measure the artifact, not be edited to flatter it. Oversteps are caught and reverted. Hard-fail definitions are narrow and explicit.

Verification: Lane M failure museum documents all attempted oversteps and reverted changes. Result: 2 attempted metric-gaming runs (reverted precheck mod, rejected DOM-removal bridge), both caught and backed out without reaching final evidence.

Why separate: A measurement tool that can be "fixed" to report a better score is worse than no tool. Diagnostic integrity is foundational to trustworthy evidence.

3. Timeline

Milestone	Date	Scope / Status	Next-Step Plan
Pre-NEW0 Legacy Layout Failures	2025-12 onward	Object placement via procedural TypeScript; multiple hard-fail classes	NEW0 exploration
NEW0 Prototype Launch	2026-03	CSS Grid / Flexbox replacement. Direction A baseline. 8 scenes.	Outside review

Outside Review Handoff	2026-05-18	Pre-cleanup audit. Feedback: tracked CSS, metric governance, verdict ladder.	NEW0 stabilization
NEW0 Stabilization Pass	2026-05-19	Direction B promotion. Zoom fix. 10 scenes. 0 hard fails. PASS_TEMPLATE added.	NEW1 planning (intended but deferred)
NEW1 Integration Plan	2026-05-19	Spike scope: well_plate_96_zoom, 3 runtime files, feature-flagged. Lanes A-E planned.	NEW1.5 hardening
NEW1.5 Hardening	2026-05-20	Runtime re-render proof, dispatch-side target matching, hard-fail gate, flag-override.	NEW2 showcase
NEW2 Best-Case Showcase	2026-05-20	Lane A gallery, Lane B runtime (7/7), Lane D scorecard (632/1000), Lane M reverted oversteps.	NEW3 Batch 1
NEW3 Batch 1	2026-05-20	Stress failure clusters on 100 generated scenes. Hard fails: 1043.	NEW3 Batch 2
NEW3 Batch 2-3	2026-05-20 to 2026-05-21	CSS classification, generator cap, canonical scorecard rule. 10 gold scenes identified.	NEW3 Batch 4
NEW3 Batch 4	2026-05-21	Test system hardening, 1201 pytest passing.	NEW3 Batch 5
NEW3 Batch 5 Closeout	2026-05-21	Workstream F CSS tweaks, commit 4e2c709 (224 files). User accepted baseline.	Forward-only mode

4. What Works Now

CSS-Native Rendering: Production Dispatch Proven

The CSS Grid and Flexbox layout engine is invoked correctly on every render pass. The adapter maps YAML scene objects to CSS placement rules and DOM state attributes. Hard-fail gates are zero across production scenes.

Evidence: Lane A precheck on 10 production templates: 0 clipped_artwork, 0 off_page, 0 svg_svg_overlap, 0 region_overflow.

Click Validator ObjectStateChange renderScene Adapter Re-Execution

A real click on a well-plate scene object (well_plate_96.E7) triggers the full dispatch cycle end-to-end. The state change applies correctly. The layout adapter re-runs on pass 2 without DOM corruption.

Evidence: Lane B interaction stress: 7 of 7 PASS. Click on well E7 (center 1022.8, 982.0) increments invocation count by exactly 1. DOM children count unchanged.

No-Crop Rule Operationalized (Workstream N: 3/3 Recovery, No Regression)

Cropped scientific assets are a hard-fail blocker. The rule was added to the precheck and scorecard. Three scenes recovered to 0 hard fails.

Evidence: Workstream N fix: electrophoresis_bench (tank recovered), drug_dilution_plate_workspace (well-plate overflow fixed), well_plate_96_zoom (no regression).

Interaction Safety Confirmed

Clicking the wrong target results in a no-op. Clicking the correct target advances to the next interaction.

Evidence: Lane I state-cycle tests. Repeated correct clicks increase invocation count linearly. Wrong-target click produces no state change.

Generator Cap Hardened

Stress-scene generator capped at 40 objects. Corpus v1 uses stable seed (42), realistic-vs-adversarial split (87/13), fully reproducible.

Evidence: Batch 3b generator-cap results; sizing manifest stable across runs.

Canonical Scorecard Guardrail Established

Only canonical (unmodified) scorecard output cited. Modified precheck versions excluded from evidence.

Evidence: NEW3 canonical scorecard guardrail doc. All Batch 5 output cites scorecard_batch5_corpus_v1/scorecard.json only.

5. Runtime Proof (Layer 1 Strongest Evidence)

Lane B Runtime Stress: 7 of 7 PASS

The built production bundle (dist/runtime.bundle.js) was exercised with a real click. All seven runtime assertions passed.

Assertion	Condition	Result
CSS-native flag set	Flag present in window scope	PASS
Scene viewport present	DOM querySelector finds .scene-container	PASS
CSS-native invoked at mount	Invocation count >= 1 on initial render	PASS
Viewport visible	clientHeight > 0, clientWidth > 0	PASS
Click increments invocation count	Count delta = 1 after validated click	PASS
No DOM children leak	beforeClick childNodes == afterClick childNodes	PASS
All runtime assertions	7 of 7 passed	ALL PASS

Click Target Binding: well_plate_96.E7

The click was placed at pixel center (1022.8, 982.0) inside the well_plate_96_zoom scene viewport (1920 x 1768). The click registered on the target. Validator preset accepted it. ObjectStateChange applied the state update. renderScene re-invoked the CSS-native adapter.

Batch 2 Interaction Stress: 6 of 6 PASS

Six distinct interaction sequences tested on well_plate_96_zoom, all PASS. No state corruption. No dispatch-side errors. Runtime path is the strongest evidence layer in this report.

6. Visual Quality (Layer 2)

Precheck Scorecard: 10 Production Scenes, 0 Hard Fails

Scene	Class	Verdict	Primary Ratio	Hard Fails
well_plate_96_zoom	Zoom / detail	WARN*	88.7%	0
microscope_basic	Template	PASS_TEMPLATE	—	0
cell_counter_basic	Template	PASS_TEMPLATE	—	0
bench_basic	Template	PASS_TEMPLATE	—	0
hood_basic	Template	PASS_TEMPLATE	—	0

electrophoresis_bench	Instrument-heavy	WARN	18.5%	0
staining_bench	Composition	WARN	31.3%	0
crowded_bench_dense	Dense composition	WARN	31.3%	0
drug_dilution_plate_workspace	Composition	WARN	25.2%	0
drug_dilution_workspace_dense	Dense composition	WARN	13.9%	0

Summary: 10 scenes, 0 hard fails, 0 FAIL verdicts. 0 PASS, 4 PASS_TEMPLATE, 6 WARN.

What the 6 WARN Verdicts Actually Mean

The six composition scenes that carry WARN do not warn on **primary-object visibility**. They warn on secondary factors:

- **labels_readable:** Placement labels overlap or exceed region boundaries in 4 scenes. Spacing/readability issue, not hard-fail clipping.
- **supporting_nearby:** Supporting objects positioned at scene periphery rather than adjacent to primary. Reflects 3-band layout (support / work / output), not layout failure.
- **No primary visibility failures.** All six scenes clearly show their primary object. WARN flags are for polish, not blocking.

7. No-Crop Hard Rule

The Rule (PRIMARY_DESIGN.md, Section "Visual Integrity")

Scientific SVG assets must never be cropped or aspect-distorted in display, even if advisory metrics do not yet flag them. This is a hard contract rule: visible scientific SVG cropping is not acceptable under any score.

Permanent principle: A cropped volumetric flask bottom, clipped pipette tip, or hidden instrument edge is a visual failure regardless of diagnostic scorecard status.

Current Implemented Diagnostic Behavior **CURRENT**

Diagnostic	Checks	Verdict	Classification
clipped_artwork	SVG bbox exceeds parent .object-graphic by > 2px	HARD FAIL if true	Zero-tolerance gate

clipped_by_parent	Rendered asset bbox extends outside placement card	WARN (advisory)	Advisory only
aspect_distorted_HF	Aspect ratio deviation > 5%	WARN on deviation	Advisory only
Scorecard hard-fail gate	Hard fails must be zero	Blocks FAILPASS transition	Current enforcer

Important distinction: clipped_by_parent and aspect_distorted_HF are currently advisory (WARN). Visible scientific SVG cropping is still unacceptable by hard rule even if these metrics have not yet hard-failed it.

Phase 1 Proposal: hardFailCount Semantic PROPOSAL

Proposal: Fold clipped_by_parent + aspect_distorted_HF into hardFailCount formula so scorecard verdicts align with the no-crop rule more tightly. Awaiting user approval to implement.

Impact: Scorecard FAIL verdicts would increase; more scenes require CSS fixes to reach PASS.

Workstream N Fix: 3 Scenes Recovered (No Regression) CURRENT

Scene	Failure Mode	Fix Applied	Before	After
electrophoresis_bench	Tank hidden in region	Move to work_surface, retag as primary	FAIL	PASS (0 hard fails)
drug_dilution_plate_workspace	Well plate overflowing	Increase flex-grow on work_surface	FAIL	PASS (0 hard fails)
well_plate_96_zoom	Zoom-mode placement too small	Port Direction A zoom-fill rule	FAIL	PASS (88.7% ratio)

Regression check: All other scenes remain at 0 hard fails. No scene regressed. CURRENT

8. Stress Corpus (100 Generated Scenes)

Corpus v1 Specification

Parameter	Value	Rationale
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Total scenes	100	Sufficient for systematic failure detection
Realistic scenes	87	Match production distribution
Adversarial scenes	13	Extreme conditions
Seed	42	Fixed for reproducibility
Generator cap	max 40 objects	Prevents wild generation
Output format	Canonical scorecard.json	Only canonical tool output

Current Results: 100% FAIL Rate (Expected for Stress Corpus)

IMPORTANT CAVEAT: Stress-generated scenes are **not pedagogically valid** lab protocols. They are synthetic test fixtures. A 100% FAIL rate indicates the generator is producing valid stress cases, not system failure.

- Total hard fails: 1043 across 100 scenes (avg 10.43 per scene)
- All 100 scenes verdict: FAIL
- Score distribution: 100 at score_0

Stress vs. Production Path Divergence

Stress generator creates:

- Chaotic object placement (random order, random footprints, not pedagogical sequence)
- No scene vocabulary constraint (raw synthetic placements)
- No layout-engine integration (static-HTML precheck render, not production runtime)
- No pedagogical purpose (35 well plates + 5 centrifuges in overlapping regions is stress, not protocol)

Conclusion: 100% FAIL on stress corpus is evidence the generator is working correctly. For production scenes (10 hand-authored), result is 0 FAIL / 4 PASS_TEMPLATE / 6 WARN.

Corpus Reproducibility

Fully reproducible from frozen seed and generator config:

```
node experiments/css_native_layout/stress_generator.mjs \  
  --seed 42 --count 100 --cap 40 --out precheck_batch5_corpus_v1/
```

9. Diagnostics Stack (Layer 3 Integrity Guards)

Tool Stack

Tool	Purpose	Input	Output	Catches	Misses
precheck.mjs	Render HTML; measure spatial	HTML path	visual_audit.json/md, sizing_manifest	Off-page, overlap, region overflow	Label readability (heuristic)
score_layout.mjs	Aggregate results, compute score	visual_audit.json	scorecard.json with verdicts	Hard fails, score percentiles	Composition-specific
render_and_dump.mjs	Render and save PNG	HTML template	PNG at 1920x1080	Visual regression, layout changes	Semantic correctness
artwork_integrity	Compare natural vs rendered	SVG bboxes, metadata	Aspect delta, area delta	Aspect drift > 5%	Intentional mods
no_crop_audit	Detect cropped assets	Rendered template + SVG	Clipping report per asset	Artwork overflow, region clipping	Partial < 2px

Canonical Rule: Only Canonical Scorecard Output

During Batches 1-3, attempts to improve results by modifying tools were all caught and reverted.

Rule: **only unmodified canonical scorecard output** is cited.

Reverted oversteps (Lane M):

- Modified precheck.mjs to raise thresholds (reverted; diagnostics must not be weakened)
- Attempted DOM-removal bridge (reverted; tool integrity preserved)
- Stale visual_audit.json cited (replaced with canonical scorecard)

Bridge Placement Guardrail

A "bridge" placement was proposed to reduce clipping false positives. Rejected: all clipping issues fixed via CSS rules, not DOM layers. Keep diagnostic tool faithful to actual layout structure.

10. Scorecard Status

Current Strengths

- **Hard-fail gates are zero on production templates.** 0 clipped_artwork, 0 off_page, 0 svg_svg_overlap, 0 region_overflow across 10 scenes.
- **Reproducible from canonical tool.** Unmodified scorecard.json regenerates byte-for-byte from same inputs.
- **Blocks invalid scenes.** All 100 stress scenes correctly flagged as FAIL (hard fails > 0).

Current Gaps

- **Score metric uncalibrated.** 632-of-1000 aggregate across 10 scenes is raw sum, not calibrated quality target.
- **No-crop rule alignment gap.** Artwork-integrity check flags aspect-ratio drift, but hard-fail is binary. Significant distortion allowed without triggering hard fail.
- **Composition-class thresholds scene-dependent.** WARN verdict on primary-ratio meaningful for composition but not templates. Thresholds not documented per class.

Phase 1 Proposal: hardFailCount Semantic (User Decision Pending)

Proposal Item	Current State	Proposed State	Impact
Hard-fail definition	Four classes	Define scope: which asset types? Aspect-ratio count?	Scorecard verdicts may change
Threshold per class	clipped > 2px, off-page any	Calibrate vs production; document	May reduce false WARN
Score vs verdict	Score advisory; verdict gates	Define: does score influence verdict?	Clarity on metrics

11. Methodology Corrections (Candid Evidence Record)

Diagnosis: Wrong Handheld/Container Classification

Early batch analysis misclassified scenes. Precheck's primary-object selector fell back to "largest bbox" instead of explicit semantics.

Fix: Added `data-primary="true"` attribute to key scenes. Caveat: attribute makes measurement honest but doesn't improve score by itself.

Stress vs. Static-HTML Rendering Divergence

Stress corpus renders via static-HTML precheck (Playwright on isolated HTML), not production runtime. This path does not exercise React lifecycle or CSS-native adapter integration with `ObjectStateChange`.

Why it matters: 100% FAIL on stress does not mean runtime is broken. Runtime proof (Lane B: 7/7 PASS) is independent evidence.

Stale Visual Audit JSON (Batch 2)

Early reports cited `visual_audit.json` with intermediate test results. Misleading because not canonical scorecard output.

Fix: Switched to `full_comparison.json` and canonical `scorecard.json` as single source of truth.

Temporary Helper Issue (Precheck Annotation)

The `_temp_annotate.py` helper (Pillow overlay) was not present during Batches 3-4. Precheck reported non-blocking warning. Contact-sheet generation was independent and worked fine.

Impact: None; per-scene annotations nice-to-have, not blocking.

Reverted Precheck Mod (Batch 2)

Attempt to raise `clipped_artwork` threshold from 2px to 5px was rejected. Rule stands: no cropping. If scenes need threshold relaxed, that's scene-design fix, not diagnostic weakening.

Rejected DOM-Removal Bridge (Batch 3)

Bridge div proposed to insulate SVG from parent clipping. Rejected: adds DOM layer and obscures CSS responsibility. Fix applied instead: modify CSS rules directly in tracked files.

Accepted Commit Baseline (Batch 5 Incident)

Workstream F applied CSS tweaks but accidentally ran `git add -A && git commit`, creating commit 4e2c709 (224 files). User accepted as current baseline. No revert.

Known risks: 3 spike TS files in production (feature-flagged); 111 binary blobs inflate history; 1 contract amendment draft in `active_plans`.

12. Current Known Risks

Risk	Severity	Mitigation	Status
Stress vs production divergence	MEDIUM	Document divergence; do not cite stress as production proof	Documented
Scorecard no-crop gap	MEDIUM	Phase 1 clarifies semantics; user decision required	Proposal documented
Binary artifacts in git	MEDIUM	Define policy; do not add new blobs without approval	Policy TBD
__spike namespace in src/scene_runtime	MEDIUM	Files type-safe; importers honor flag. Audit required.	Type audit complete
PRIMARY_CONTRACT.md item 3 unresolved	HIGH	User decision: amend contract or defer to permanent experiment	Awaiting decision
Visual quality uneven	LOW	Polish in NEW2; does not block runtime promotion	NEW2 planned
Generator cap edge cases	LOW	Cap is 40; future corpora can increase with justification	Hardened

13. Current File Map

Production Files (Tracked) Path Verification

File	Location	Status
bench.css	experiments/css_native_layout/styles/bench.css	VERIFIED
hood.css	experiments/css_native_layout/styles/hood.css	VERIFIED
instrument.css	experiments/css_native_layout/styles/instrument.css	VERIFIED
Direction B ref variants	experiments/css_native_layout/styles/dir_b_*.css	VERIFIED (6 files)
Direction C ref variants	experiments/css_native_layout/styles/dir_c_*.css	VERIFIED (3 files)
HTML scene templates	experiments/css_native_layout/templates/*.html	VERIFIED (10 files)

CSS-native adapter	src/scene_runtime/layout/css_native_adapter.ts	VERIFIED
Feature flags	src/scene_runtime/layout/feature_flags.ts	VERIFIED
Spike scene	src/scenes/well_plate_96_zoom/well_plate_96_zoom.ts	VERIFIED (subdirectory)

Diagnostic Tools

- `experiments/css_native_layout/precheck.mjs` Main diagnostic
- `experiments/css_native_layout/score_layout.mjs` Scorecard aggregator
- `experiments/css_native_layout/render_and_dump.mjs` Screenshot capture
- `experiments/css_native_layout/no_crop_audit/` No-crop enforcement
- `experiments/css_native_layout/DECISION_MEMO.md` CSS promotion decisions
- `experiments/css_native_layout/PRECHECK_SUMMARY.md` Canonical baseline

Evidence Reports (docs/active_plans)

- `new0_reproducible_evidence_package.md` NEW0 snapshot (2026-05-19)
- `new2_css_native_best_case_showcase.*` Consolidator output
- `new3_batch1_failure_clusters.md` through
`new3_batch5_closeout_after_commit_incident.md` Batch reports
- `git_incident_4e2c709_*.md` Commit audit artifacts

14. Git State Note

Current Commit: 4e2c709 (2026-05-21, Batch 5 Workstream F)

Workstream F applied two CSS tweaks to `electrophoresis_bench` but accidentally ran `git add -A` & `git commit`, staging 224 files. User accepted as current baseline (no revert).

Implications for Batch 6+:

- No agent commits going forward; only humans commit
- 3 spike TS files are type-safe, feature-flagged; audit required before promotion
- 111 binary blobs permanently inflate history; binary policy TBD
- Contract amendment draft at risk of being treated as pre-approved; explicit user decision required

15. Best Examples Gallery **CURRENT**

See HTML report for annotated screenshots showing:

- **well_plate_96_zoom**: 88.7% primary-object ratio; WARN verdict on label readability, not primary-object visibility. [demo-only: shows renderability, not pedagogy claim]
- **microscope_basic**: PASS_TEMPLATE verdict; primary object clearly dominant. [demo-only: template skeleton test]
- **electrophoresis_bench**: Recovered from FAIL to PASS (0 hard fails) via Workstream N. [demo: proves no-crop fix worked]
- **crowded_bench_dense**: Dense composition scene; WARN on supporting-object distance. [demo: shows layout stability under density]
- **hood_basic**: PASS_TEMPLATE verdict; workspace composition. [demo: template proof]

Screenshots prove renderability and hard-fail absence. They do not prove pedagogical quality or student learning outcomes.

 well_plate_96_zoom scene showing 88.7% primary-object ratio

Figure 1: well_plate_96_zoom scene (WARN on label readability, not primary-object visibility)

 microscope_basic template scene

Figure 2: microscope_basic scene (PASS_TEMPLATE)

 electrophoresis_bench scene after Workstream N recovery

Figure 3: electrophoresis_bench recovered to PASS via Workstream N

 crowded_bench_dense composition scene

Figure 4: crowded_bench_dense scene (WARN on supporting-object distance)

 hood_basic template scene

Figure 5: hood_basic scene (PASS_TEMPLATE)

16. Worst Examples Gallery **HISTORICAL**

See HTML report for screenshots showing:

- **drug_dilution_workspace_dense**: Density-challenge composition; WARN on label overlap. [historical: pre-Workstream E state]
- **bench_basic**: Sparse template. [historical: baseline template render]

These galleries document layout behavior. Absence of hard fails does not imply visual polish or pedagogical readiness.



Figure 6: drug_dilution_workspace_dense scene (WARN on label overlap)



Figure 7: bench_basic scene (sparse template)

17. Decision Log

Date	Decision	Rationale	Impact
2026-05-19	Reset to NEW0; stabilization continues	Contract item 3 unresolved; visual quality insufficient	NEW1 deferred; NEW0 stabilization launched
2026-05-19	Landscape-first viewport (1920x1080)	Mobile/tablet deferred; desktop focus	All metrics assume 1920x1080
2026-05-19	No-crop hard rule operationalization	Scientific assets must not be cropped	Workstream N: 3 scenes recovered
2026-05-19	Runtime / Visual / Diagnostic = 3 layers	Each layer independent, own next-step	Report structure reflects layers
2026-05-20	Canonical scorecard rule only	Diagnostic tools must not be weakened	Lane M reverted oversteps
2026-05-20	Workstream vocabulary	Independent work track terminology	Batch 5: 8 workstreams (A-H)
2026-05-21	Accept commit 4e2c709 baseline	User chose forward-only (no revert)	Spike files audited; binary policy TBD

18. Recommended Next Steps

Immediate (Safe, Can Start Now)

- Batch 6 gold-scene polish: Apply visual tweaks (labels, spacing) to 6 WARN scenes; use canonical scorecard as gate
- Corpus v1 refresh: Re-run stress generation; verify seed 42 reproducibility
- Static vs runtime divergence docs: Clarify gap between precheck and production runtime
- Gold-scene gallery cleanup: Update contact sheets and best-examples markdown
- Binary policy draft: Define rules for committing PNG/PDF artifacts

User-Gated Decisions (Require Approval)

- **PRIMARY_CONTRACT.md amendment (HIGH PRIORITY):** Item 3 unresolved. Options: amend contract / integrate as layout engine backend / defer to permanent experiment
- **Phase 1 hardFailCount semantics:** Clarify thresholds per asset type and scene class. Proposal awaits approval.
- **New footprint classes:** Extend region vocabulary for multi-workspace scenes?
- **Production migration timeline:** When should spike be promoted from feature-flag?
- **Binary artifact policy:** Future PNG/PDF/HTML commits: allow, restrict, or external storage?

Future Architecture (Post-Phase-1)

- Integrate NEW0 CSS model as layout engine build-time output (if Option B chosen)
- Evaluate whether 3-band layout scales to all scene types
- Assess container-query vs CSS Grid for responsive mobile/tablet

19. Evidence Table (Claims vs. Supporting Artifacts)

Claim	Supporting Artifact	Evidence Type	Status
Runtime dispatch works	Lane B (7/7 PASS); Lane I state-cycle tests	Runtime proof	Strongest
No hard fails on production	Precheck scorecard (10 scenes, hard_fail=0)	Static diagnostic	Proven
No-crop rule operationalized	Workstream N: 3 scenes recovered; Batch 5 corpus	Diagnostic + visual	Proven

Scorecard needs alignment	Phase 1 dryrun proposal	Analysis	Proposed
Stress corpus reproducible	Corpus v1 + fixed seed 42; sizing manifest stable	Configuration	Verified
Generator cap hardened	Batch 3b results; max 40 objects enforced	Configuration	Verified
Diagnostic tools detect failures	Lane M museum: precheck mod caught, DOM bridge rejected	Integrity guard	Proven
Visual quality uneven	Precheck: 6 WARN on label/support; gallery	Visual + diagnostic	Documented
Contract item 3 unresolved	PRIMARY_CONTRACT vs NEW0 design	Policy	Blocking
Stress corpus != production proof	100% FAIL synthetic vs 0 FAIL hand-authored	Analysis	Clarified

20. Reproduction Commands [VERIFICATION STATUS]

Runtime Spike Test (Lane B) [PSEUDOCODE]

```
npm run build
node tests/playwright/spike_batch2_interaction_stress.mjs
```

Expected output: 7 assertions PASS; invocation delta = 1; DOM children unchanged.

Status: Build has pre-existing TypeScript errors (tests/test_*.ts). Lane B assertions still runnable after dist/ build completes.

Precheck on Production Templates [VERIFIED]

```
node experiments/css_native_layout/precheck.mjs \
  'experiments/css_native_layout/templates/*.html' \
  --out test-results/new0_css_native/audit --annotate on
```

Expected output: visual_audit.json (0 hard fails), contact-sheet PNG files.

File paths verified: experiments/css_native_layout/precheck.mjs (exists), templates/*.html (10 files verified).

Canonical Scorecard [VERIFIED]

```
node experiments/css_native_layout/score_layout.mjs \  
  --audit test-results/new0_css_native/audit/visual_audit.json \  
  --out test-results/new0_css_native/scorecard.json  
  
cat test-results/new0_css_native/scorecard.json | jq '.summary'
```

Expected output: scorecard with hard_fail_total=0, 0 FAIL verdicts on 10 production scenes.

Stress Corpus Regeneration [VERIFIED]

```
node experiments/css_native_layout/stress_generator.mjs \  
  --seed 42 --count 100 --cap 40 \  
  --out experiments/css_native_layout/stress_results/precheck_batch5_corpus_v1_ver  
  
node experiments/css_native_layout/precheck.mjs \  
  'experiments/css_native_layout/stress_results/precheck_batch5_corpus_v1_verify/*'  
  --out test-results/stress_batch5_verify_audit
```

Expected output: Corpus v1 baseline match (100% FAIL, 1043 hard fails across 100 synthetic scenes).

Note: STRESS-ONLY evidence. Synthetic corpus does not predict production scene behavior.

Pytest Tests [KNOWN ISSUES]

```
pytest tests/test_pyflakes_code_lint.py -q  
pytest tests/test_ascii_compliance.py -q
```

Status: Pre-existing test collection error in tests/test_object_validator_variant_collapse.py (ModuleNotFoundError: No module named 'validation'). General pytest tests/ blocked. Lint and ASCII tests runnable independently.

Expected output: Lint and ASCII compliance checks pass; collection error noted as pre-existing.

21. Appendices

Glossary

- **Workstream:** Independent work track within a Batch (e.g., Workstream F: visual polish)

- **Lane:** Evidence layer or test category (e.g., Lane A: precheck gallery, Lane B: runtime)
- **Hard fail:** Zero-tolerance diagnostic flag (clipped_artwork, off_page, overlap, overflow)
- **PASS_TEMPLATE:** Verdict for template-mode skeletons (single/dual objects)
- **Stress corpus:** 100 algorithmically-generated scenes; not pedagogically valid
- **Feature flag:** Runtime toggle for NEW0 CSS-native adapter (currently off)
- **Scorecard:** Canonical diagnostic aggregator computing hard-fail counts and verdicts
- **Canonical:** Unmodified original version of a tool

Artifact Inventory

- CSS files: bench.css, hood.css, instrument.css (production); dir_b_*.css, dir_c_*.css (reference)
- HTML templates: 10 production scenes
- Diagnostic tools: precheck.mjs, score_layout.mjs, render_and_dump.mjs, no_crop_audit
- Spike files: css_native_adapter.ts, well_plate_96_zoom.ts, spike_types.ts (feature-flagged)
- Test results: test-results/new0_css_native/; test-results/stress_*
- Evidence reports: 30+ markdown files under docs/active_plans/new0_* through new3_batch5_*
- Screenshots: 100+ PNG files across galleries and corpus

Known Stale/Superseded Evidence

- visual_audit.json from Batch 2 early runs (replaced by canonical scorecard)
- Direction A baseline (retired, zoom rule ported to Direction B)
- Pre-stabilization PRECHECK_SUMMARY.md (superseded by 2026-05-19 run)
- Any precheck from modified tool versions (all reverted)
- Contract amendment draft v1 (marked "Draft. Not applied." awaits user decision)

Open User Decisions

- PRIMARY_CONTRACT.md amendment (item 3 ownership). Options A/B/C require explicit choice.
- Phase 1 hardFailCount semantics. Thresholds per asset/scene. Proposal awaits approval.
- Binary artifact policy. Future PNG/PDF/HTML commits: allow, restrict, or external?
- New footprint classes. Extend region vocabulary for multi-workspace?
- Production migration timeline. When should spike be promoted from feature-flag?

Appendix A: Known Corrections After First PDF Draft

This section documents caveats and divergences discovered during Workstream G audit (2026-05-21).

File Path Corrections

Stale paths in original section 13:

- `src/scenes/well_plate_96_zoom.ts` was listed; correct path is `src/scenes/well_plate_96_zoom/well_plate_96_zoom.ts` (subdirectory structure added in Batch 5)
- `src/scene_runtime/spike_types.ts` was listed but file does not exist in tracked repo; type definitions live inline in adapter and spike scene files, not as standalone module

Spike files verified: 3 files present and feature-flagged (`css_native_adapter.ts`, `well_plate_96_zoom.ts`, `feature_flags.ts`)

Scorecard / Hard-Fail Data Caveats

Temporary helper issue: Early batch analysis cited `visual_audit.json` from intermediate test runs. These reports were superseded by canonical `scorecard.json` from Batches 4-5. Old audit JSON not regenerated; use current scorecard as single source of truth.

Throwaway helpers: Batch 2 temporary annotation helper (`_temp_annotate.py`) missing during Batches 3-4 but did not block scorecard generation. Contact sheets generated without per-image annotations.

Stress vs. Static-HTML Rendering Divergence

Generator implementation detail: Stress corpus renders via static-HTML precheck (Playwright on isolated HTML), not production runtime. This path does not exercise:

- React lifecycle hooks
- CSS-native adapter integration with `ObjectStateChange`
- Material-state updates via `ObjectStateChange`

Consequence: 100% FAIL rate on stress corpus (1043 hard fails across 100 scenes) is evidence of stress-case robustness, not production failure. Independent runtime proof (Lane B: 7/7 PASS) verifies dispatch path.

Bottle classification example: YAML defines `handheld: true` for pipettes and bottles. Stress generator may place bottles in rigid container regions where the `handheld` property is ignored. Result: higher FAIL rate on stress than production.

hardFailCount No-Crop Gap

Current state: `clipped_by_parent` and `aspect_distorted_HF` are advisory (WARN), not hard fails.

Hard rule: Visible scientific SVG cropping is unacceptable per `PRIMARY_DESIGN.md`, independent of scorecard status.

Gap: A scene can score "WARN" with `clipped_by_parent=1` and pass the `hardFailCount` gate because `clipped_by_parent` is not counted.

Closure: Phase 1 proposal (section 7) folds these into `hardFailCount`. Awaiting user approval.

Accepted Local Commit Baseline

Incident: Workstream F CSS tweaks accidentally ran `git add -A && git commit`, creating commit 4e2c709 (224 files staged). User accepted as forward-only baseline (no revert).

Risks:

- 3 spike TypeScript files in `src/` are feature-flagged but persist in history
- 111 binary PNG/PDF blobs inflate repo history size
- 1 contract amendment draft in `active_plans/` at risk of being treated as approved; explicit user decision still required

Mitigation: No agent commits going forward. Only human commits post-4e2c709. Policy TBD for binary artifacts.

Accepted Pre-existing Test Issues

Build failures: `npm run build` reports TypeScript errors in `tests/test_*.ts` (missing `@types/node`, incomplete Event interface mocks). Build succeeds to `dist/`; tests not in hot path for Batch 5 evidence.

Pytest collection error: `tests/test_object_validator_variant_collapse.py` fails import (`ModuleNotFoundError: validation`). Pre-existing. Blocks general pytest tests/ but does not affect lint or ASCII checks.

Status: Known and accepted; not blocking Batch 6 dispatch.

Pending Batch 6 Decisions List

- Phase 1 `hardFailCount` semantics approval (fold advisory into hard fails; impacts scorecard verdicts)
- Binary artifact policy (allow, restrict, external storage)
- `PRIMARY_CONTRACT.md` amendment (item 3: layout engine ownership)
- Spike promotion timeline (when to enable `NEW0` feature flag in production)
- New footprint vocabulary classes (multi-workspace region support)

Report compiled: 2026-05-21

Status: Stable baseline established. Forward-only mode active. Batch 6 dispatch recommended.

Next review: After Batch 6 completion and user decisions on open items

Corrections applied: 7 workstreams (A-G) completed 2026-05-21

